KAKSHA PRACTICE PAPER · CLASS 10

Mathematics (Standard) Mock Test 1

Time allowed: 3 hours **Maximum marks: 80**

Five sections A–E. Section A: 20 one-mark questions (MCQs and assertion–reason). Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks each. Section E: three case-study questions of 4 marks.

Section A

Multiple-choice questions (Q1–18) and assertion–reason questions (Q19–20). Choose the correct option.

1. The HCF of 96 and 404 is **[1]**

(a) 4 (b) 8
(c) 12 (d) 2

2. If one zero of $2x^2 - 3x + k$ is the reciprocal of the other, then k equals **[1]**

(a) 2 (b) −2
(c) 1/2 (d) 3

3. The pair of equations $2x + 3y = 5$ and $4x + 6y = 15$ has **[1]**

(a) a unique solution (b) infinitely many solutions
(c) no solution (d) exactly two solutions

4. The quadratic equation $2x^2 - 4x + 3 = 0$ has **[1]**

(a) two distinct real roots (b) two equal real roots
(c) no real roots (d) more than two real roots

5. The 10th term of the AP 2, 7, 12, … is **[1]**

(a) 45 (b) 47
(c) 52 (d) 42

6. The distance between the points (2, 3) and (4, 1) is **[1]**

(a) 2 (b) $2\sqrt{2}$
(c) 4 (d) $\sqrt{2}$

7. The midpoint of the segment joining (−1, 7) and (3, −3) is **[1]**

(a) (1, 2) (b) (2, 1)
(c) (1, 5) (d) (2, 4)

8. If $\sin A = 3/5$ and A is acute, then $\cos A$ equals **[1]**

(a) 3/4 (b) 4/5

(c) 5/4 (d) 5/3

9. $(1 + \tan^2 A)(1 - \sin A)(1 + \sin A)$ equals **[1]**

(a) 0 (b) 1

(c) $\sec^2 A$ (d) $\tan^2 A$

10. If the height of a pole and the length of its shadow are equal, the Sun's angle of elevation is **[1]**

(a) 30° (b) 45°

(c) 60° (d) 90°

11. Two tangents PA and PB from P to a circle with centre O are inclined at 80°. Then $\angle POA$ is **[1]**

(a) 50° (b) 40°

(c) 60° (d) 70°

12. The area of a sector of angle 90° in a circle of radius 14 cm is (use $\pi = 22/7$) **[1]**

(a) 616 cm^2 (b) 308 cm^2

(c) 154 cm^2 (d) 77 cm^2

13. The perimeter of a semicircular protractor of radius 7 cm is (use $\pi = 22/7$) **[1]**

(a) 22 cm (b) 29 cm

(c) 36 cm (d) 44 cm

14. The volume of a hemisphere of radius 3 cm is **[1]**

(a) 9π cm^3 (b) 18π cm^3

(c) 27π cm^3 (d) 36π cm^3

15. The mean of $x, x + 2, x + 4, x + 6$ and $x + 8$ is 11. The value of x is **[1]**

(a) 5 (b) 7

(c) 9 (d) 11

16. The empirical relationship between mean, median and mode is **[1]**

(a) Mode = 3 Median – 2 Mean (b) Mode = 2 Median – 3 Mean

(c) Median = 3 Mode – 2 Mean (d) Mean = 3 Median – 2 Mode

17. The probability that a leap year has 53 Sundays is **[1]**

(a) 1/7 (b) 2/7

(c) 3/7 (d) 5/7

18. A card is drawn at random from a well-shuffled deck of 52 cards. The probability that it is a red king is **[1]**

(a) 1/13
(b) 1/26
(c) 1/52
(d) 1/4

19. Assertion (A): √2 is an irrational number. Reason (R): The square root of every prime number is irrational. **[1]**

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

20. Assertion (A): The point (0, 4) lies on the y-axis. Reason (R): The x-coordinate of every point on the y-axis is zero. **[1]**

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Section B

Very short answer questions.

21. Prove that $3 + 2\sqrt{5}$ is irrational. **[2]**

22. Find the zeroes of the polynomial $x^2 - 3$ and verify the relationship between the zeroes and the coefficients. **[2]**

23. In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AD/DB = 3/5$ and $AC = 5.6$ cm, find AE. **[2]**

24. If $\tan A = \cot B$, prove that $A + B = 90°$ (A and B acute). **[2]**

25. Find the area of a quadrant of a circle whose circumference is 22 cm. **[2]**

Section C

Short answer questions.

26. Find the HCF and LCM of 6, 72 and 120 using prime factorisation. **[3]**

27. Solve $2x + 3y = 11$ and $2x - 4y = -24$, and hence find m for which $y = mx + 3$. **[3]**

28. Prove that the lengths of tangents drawn from an external point to a circle are equal. **[3]**

29. Prove that $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$. **[3]**

30. Find the ratio in which the y-axis divides the line segment joining (5, −6) and (−1, −4). Also find the point of division. **[3]**

31. Two dice are thrown together. Find the probability that (i) the sum is 8 (ii) a doublet appears. **[3]**

Section D

Long answer questions.

32. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less. Find the speed of the train. **[5]**

33. State and prove the Basic Proportionality Theorem. **[5]**

34. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45°. Find the height of the tower. **[5]**

35. A solid toy is a hemisphere surmounted by a right circular cone of the same radius 3.5 cm. The total height of the toy is 15.5 cm. Find its total surface area. (π = 22/7) **[5]**

Section E

Case-study questions (1 + 1 + 2 marks).

36. Case study — Auditorium seating. A school auditorium has 20 seats in the first row, 22 in the second, 24 in the third, and so on. (i) How many seats are in the 10th row? (1) (ii) If there are 30 rows, how many seats are in the last row? (1) (iii) Find the total number of seats in 30 rows. (2) **[4]**

37. Case study — Three friends' houses. On a map, Asha's house is at A(2, 1) and Bharat's house is at B(6, 4) (units in km). (i) Find the distance between the two houses. (1) (ii) A shop is exactly midway between them. Find its coordinates. (1) (iii) A bus stop divides AB in the ratio 1 : 3. Find its coordinates. (2) **[4]**

38. Case study — Pocket money survey. The daily pocket money (₹) of 50 students: 0–20: 5, 20–40: 10, 40–60: 18, 60–80: 12, 80–100: 5. (i) Find the modal class. (1) (ii) Find the median class. (1) (iii) Find the mean pocket money. (2) **[4]**

Answer key and marking points

1. (a) 4

2. (a) 2

3. (c) no solution

4. (c) no real roots

5. (b) 47

6. (b) $2\sqrt{2}$

7. (a) (1, 2)

8. (b) 4/5

9. (b) 1

10. (b) 45°

11. (a) 50°

12. (c) 154 cm^2

13. (c) 36 cm

14. (b) 18π cm^3

15. (b) 7

16. (a) Mode = 3 Median – 2 Mean

17. (b) 2/7

18. (b) 1/26

19. (a) Both A and R are true, and R is the correct explanation of A

20. (a) Both A and R are true, and R is the correct explanation of A

21. Assume $3 + 2\sqrt{5} = a/b$ where a, b are co-prime integers, $b \neq 0$. (½)
Then $\sqrt{5} = (a - 3b)/(2b)$, which is rational. (1)
This contradicts the fact that $\sqrt{5}$ is irrational, so $3 + 2\sqrt{5}$ is irrational. (½)

22. $x^2 - 3 = (x - \sqrt{3})(x + \sqrt{3})$, so the zeroes are $\sqrt{3}$ and $-\sqrt{3}$. (1)
Sum = 0 = $-b/a = -0/1$ ✓. Product = $-3 = c/a = -3/1$ ✓. (1)

23. By the Basic Proportionality Theorem, AD/DB = AE/EC = 3/5. (1)
So AE = (3/8) × AC = (3/8) × 5.6 = 2.1 cm. (1)

24. $\cot B = \tan(90° - B)$. (1)
So $\tan A = \tan(90° - B)$, giving $A = 90° - B$, i.e. $A + B = 90°$. (1)

25. $2\pi r = 22 \Rightarrow r = 22 \times 7 / 44 = 3.5$ cm. (1)
Area of quadrant = $\pi r^2/4 = (22/7 \times 3.5 \times 3.5)/4 = 38.5/4 = 9.625$ cm^2. (1)

26. $6 = 2 \times 3$, $72 = 2^3 \times 3^2$, $120 = 2^3 \times 3 \times 5$. (1)
HCF = $2 \times 3 = 6$. (1)
LCM = $2^3 \times 3^2 \times 5 = 360$. (1)

27. Subtracting the equations: $7y = 35$, so $y = 5$. (1)
Then $2x = 11 - 15 = -4$, so $x = -2$. (1)
$5 = m(-2) + 3 \Rightarrow m = -1$. (1)

28. Let PA and PB be tangents from P to a circle with centre O. Join OA, OB, OP. (½)
$OA \perp PA$ and $OB \perp PB$ (tangent $\perp$ radius), so $\angle OAP = \angle OBP = 90°$. (1)
In $\triangle OAP$ and $\triangle OBP$: OA = OB (radii), OP is common, so the triangles are congruent by RHS. (1)
Hence PA = PB (CPCT). (½)

29. LHS = $\sin^2 A + \text{cosec}^2 A + 2 + \cos^2 A + \sec^2 A + 2$. (1)
$= (\sin^2 A + \cos^2 A) + 4 + (1 + \cot^2 A) + (1 + \tan^2 A)$. (1)
$= 1 + 4 + 2 + \tan^2 A + \cot^2 A = 7 + \tan^2 A + \cot^2 A$ = RHS. (1)

30. Let the ratio be $k : 1$. x-coordinate $= (-k + 5)/(k + 1) = 0 \Rightarrow k = 5$, so the ratio is 5 : 1. (2)
$y = (5 \times (-4) + 1 \times (-6))/6 = -26/6 = -13/3$. Point = $(0, -13/3)$. (1)

31. Total outcomes = 36. (½)
(i) Sum 8: (2,6), (3,5), (4,4), (5,3), (6,2) $\Rightarrow$ P = 5/36. (1½)
(ii) Doublets: 6 outcomes $\Rightarrow$ P = 6/36 = 1/6. (1)

32. Let the speed be x km/h. $360/x - 360/(x + 5) = 1$. (1)
$360 \times 5 = x(x + 5) \Rightarrow x^2 + 5x - 1800 = 0$. (2)
$(x + 45)(x - 40) = 0 \Rightarrow x = 40$ (speed cannot be negative). (1)
Speed = 40 km/h. (1)

33. Statement: If a line is drawn parallel to one side of a triangle intersecting the other two sides at distinct points, the other two sides are divided in the same ratio. (1)
Given $\triangle ABC$ with DE ∥ BC. To prove AD/DB = AE/EC. Join BE and CD; draw $EN \perp AB$ and $DM \perp AC$. (1)
ar(ADE)/ar(BDE) = AD/DB and ar(ADE)/ar(DEC) = AE/EC (same heights). (1½)
ar(BDE) = ar(DEC) (same base DE, between the same parallels). (1)
Hence AD/DB = AE/EC. (½)

34. Let the distance between building and tower be x. Angle of depression $45° \Rightarrow \tan 45° = 7/x \Rightarrow x = 7$ m. (2)
Height of tower above the building's top = $x \tan 60° = 7\sqrt{3}$ m. (2)
Height of tower = $7 + 7\sqrt{3} = 7(1 + \sqrt{3}) \approx 19.12$ m. (1)

35. Height of cone = 15.5 − 3.5 = 12 cm. (1)
Slant height $l = \sqrt{(12^2 + 3.5^2)} = \sqrt{156.25} = 12.5$ cm. (1)
TSA = CSA of hemisphere + CSA of cone $= 2\pi r^2 + \pi r l = \pi r(2r + l)$. (1)
$= (22/7) \times 3.5 \times (7 + 12.5) = 11 \times 19.5 = 214.5$ cm². (2)

36. The seats form an AP with $a = 20$, $d = 2$.
(i) $a_{10} = 20 + 9 \times 2 = 38$. (1)
(ii) $a_{30} = 20 + 29 \times 2 = 78$. (1)
(iii) $S_{30} = 30/2 \times (20 + 78) = 15 \times 98 = 1470$ seats. (2)

37. (i) $AB = \sqrt{(4^2 + 3^2)} = 5$ km. (1)
(ii) Midpoint = (4, 2.5). (1)
(iii) Using the section formula with m : n = 1 : 3: $x = (1 \times 6 + 3 \times 2)/4 = 3$, $y = (1 \times 4 + 3 \times 1)/4 = 7/4$. Bus stop = (3, 7/4). (2)

38. (i) Highest frequency 18 $\Rightarrow$ modal class 40–60. (1)
(ii) $n/2 = 25$. Cumulative frequencies: 5, 15, 33 … $\Rightarrow$ median class 40–60. (1)
(iii) $\Sigma fx = 10\times5 + 30\times10 + 50\times18 + 70\times12 + 90\times5 = 50 + 300 + 900 + 840 + 450 = 2540$. Mean = 2540/50 = ₹50.80. (2)